6 Fractions

Tune In

Saffron, white and green are the three colours of the Indian flag. Each colour occupies the same area in the flag, which is <u>one-third</u> of the total area.

In the fraction $\frac{1}{3}$, the numerator is <u>one</u> and the denominator is <u>three</u>.

One third is the same as <u>7 out of 21</u>.

Let's Try! 1

1. $\frac{1}{7}$ $\frac{3}{5}$ $\frac{4}{9}$ $\frac{2}{7}$ $\frac{5}{7}$ $\frac{2}{6}$ $\frac{4}{7}$

2. $\frac{5}{3}$ $\frac{2}{5}$ $\frac{7}{4}$ $\frac{4}{7}$ $\frac{9}{8}$ $\frac{8}{9}$

Let's Try! 2

1. $1\frac{1}{3}$ 2. b

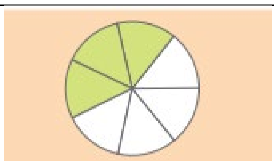
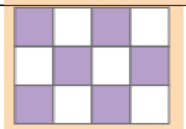
Let's Try! 3

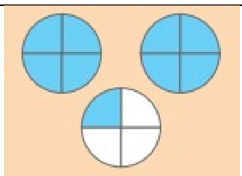
1. a. $\frac{21}{8} = 2\frac{5}{8}$ b. $\frac{32}{7} = 4\frac{4}{7}$

2. a. $2\frac{1}{5} = \frac{11}{5}$ b. $4\frac{2}{7} = \frac{30}{7}$

Exercise 6A

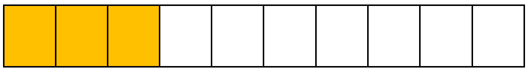
3.

b	Figure	Fraction showed by shaded part	Fraction in words
a		$\frac{3}{7}$	Three seventh
b.		$\frac{6}{12}$	One half

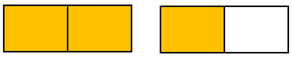
c.		$2\frac{1}{4}$	Two and one fourth

Application Based Questions

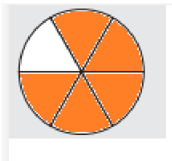
a. three tenths = $\frac{3}{10}$



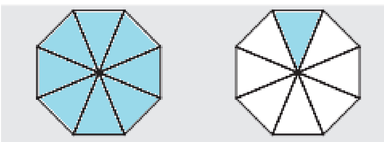
b. one and a half = $1\frac{1}{2}$



c. five sixths = $\frac{5}{6}$

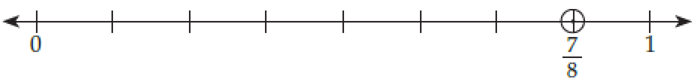


d. one and one eighth = $1\frac{1}{8}$



5. Water: $\frac{3}{4}$ Lemon syrup: $\frac{1}{4}$ sugar cubes $2\frac{1}{2}$

Let's Try! 4

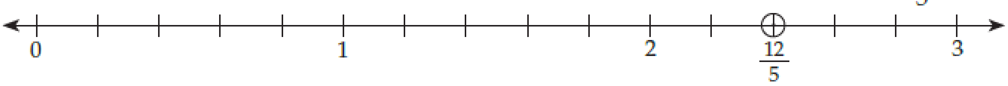
1. 

The part between 0 and 1 is divided into 8 equal parts and the 7th part is marked to represent $\frac{7}{8}$.

2. $\frac{12}{5} = 2\frac{2}{5}$, $2 < 2\frac{2}{5} < 3$

Draw a number line to show numbers 0 to 3.

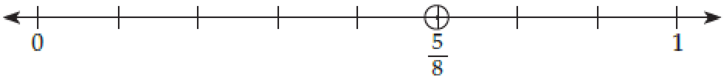
Divide each part into 5 equal smaller parts. The 12th point from 0 represents $\frac{12}{5}$.



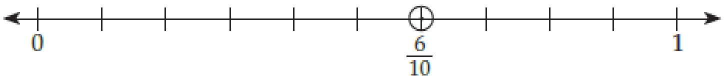
Exercise 6B

1. a.

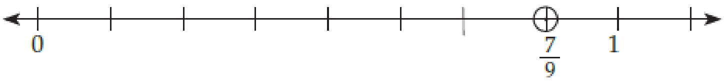
b.



c.

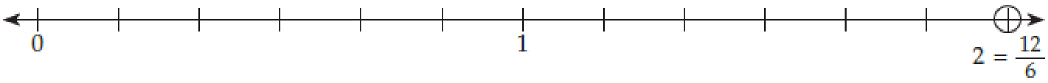


d.

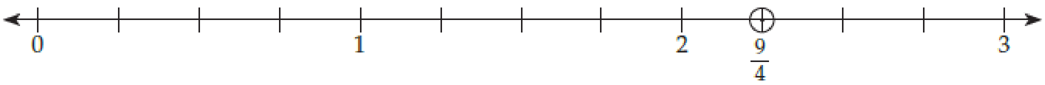


e. $\frac{8}{3} = 2\frac{2}{3}$, $\quad 2 < 2\frac{2}{3} < 3.$

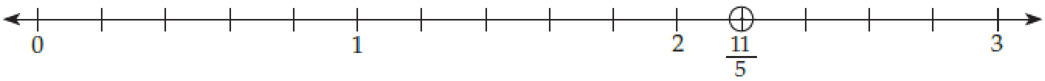
f. $\frac{12}{6} = 2$



g. $\frac{9}{4} = 2\frac{1}{4}$

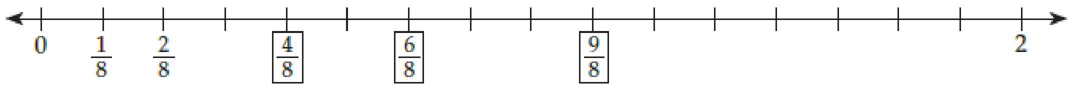


h. $\frac{11}{5} = 2\frac{1}{5}$

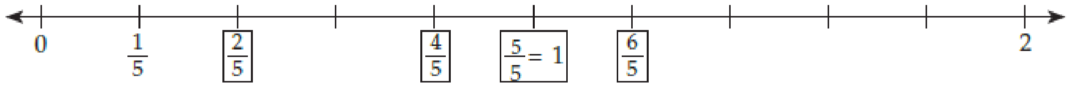


2. a. $\frac{5}{10}$ b. $\frac{3}{10}$ c. $\frac{2}{10}$ d. $\frac{9}{10}$ e. $\frac{7}{10}$

3. a.



b.



Exercise 6C

1. The given fraction is $\frac{25}{60}$.

$\frac{25}{60} = \frac{\cancel{25}^{5}}{\cancel{60}_{12}} = \frac{5}{12}$ (Dividing the numerator and denominator by 5)

$\therefore$ The answer is $\frac{5}{12}$.

2.

The given fraction is $\frac{34}{85}$.

$\frac{34}{85} = \frac{\cancel{34}^{2}}{\cancel{85}_{5}} = \frac{2}{5}$ (Dividing the numerator and denominator by 17)

$\therefore$ The answer is $\frac{2}{5}$.

3.

The given fraction is $\frac{44}{99}$.

$\frac{44}{99} = \frac{\cancel{44}^{4}}{\cancel{99}_{9}} = \frac{4}{9}$ (Dividing the numerator and denominator by 11)

$\therefore$ The answer is $\frac{4}{9}$.

4.

The given fraction is $\frac{54}{45}$.

$\frac{54}{45} = \frac{\cancel{54}^{6}}{\cancel{45}_{5}} = \frac{6}{5}$ (Dividing the numerator and denominator by 9)

$\therefore$ The answer is $\frac{6}{5}$.

5.

The given fraction is $\frac{80}{48}$.

$\frac{80}{48} = \frac{\cancel{80}^{10}}{\cancel{48}_{6}} = \frac{10}{6}$ (Dividing the numerator and denominator by 8)

$\frac{\cancel{10}^{5}}{\cancel{6}_{3}} = \frac{5}{3}$ (Dividing the numerator and denominator by 2)

$\therefore$ The answer is $\frac{5}{3}$.

6.

The given fraction is $\frac{70}{91}$.

$\frac{70}{91} = \frac{\cancel{70}^{10}}{\cancel{91}_{13}} = \frac{10}{13}$ (Dividing the numerator and denominator by 7)

$\therefore$ The answer is $\frac{10}{13}$.

Let's Try! 5

1. $\frac{6}{9} = \frac{\boxed{}}{18}$

   We know that $9 \times 2 = 18$, so we multiply the the numerator by 2.

   So, $\frac{6}{9} = \frac{6 \times 2}{9 \times 2} = \frac{\boxed{12}}{18}$

2. $\frac{5}{8} = \frac{20}{\boxed{}}$

   We know that $5 \times 4 = 20$, so we multiply the denominator by 4.

   So, $\frac{5}{8} = \frac{5 \times 4}{8 \times 4} = \frac{20}{\boxed{32}}$

Let's Try! 6

1. $\frac{8}{16}$ **and** $\frac{9}{27}$

Taking cross products, we get 8 x 27 = 216 and 16 x 9 = 144.
The cross products are not equal. So, $\frac{8}{16} \neq \frac{9}{27}$

2. $\frac{2}{7}$ and $\frac{12}{42}$

   Taking cross products we get, 7x 12 = 84 and 42 x 2 = 84.
   Therefore, the fractions are equivalent.

Exercise 6D

4. **a.** $\frac{4}{8} = \frac{1}{2}$ **b.** $\frac{6}{12} = \frac{4}{8}$ c. $\frac{2}{4} = \frac{6}{12}$ d. $\frac{1}{2} = \frac{2}{4}$

5. a. The given fraction is $\frac{10}{13}$.

   Multiplying the numerator and denominator by 2, 3, 4 and 5, we get:

   $$\frac{10 \times 2}{13 \times 2} = \frac{20}{26};\ \frac{10 \times 3}{13 \times 3} = \frac{30}{39};\ \frac{10 \times 4}{13 \times 4} = \frac{40}{52};\ \frac{10 \times 5}{13 \times 5} = \frac{50}{65}$$

   The four equivalent fractions of $\frac{10}{13}$ are $\frac{20}{26}, \frac{30}{39}, \frac{40}{52}$ and $\frac{50}{65}$.

   b. The given fraction is $\frac{1}{8}$.

   Multiplying the numerator and denominator by 2, 3, 4 and 5, we get,

   $$\frac{1 \times 2}{8 \times 2} = \frac{2}{16};\ \frac{1 \times 3}{8 \times 3} = \frac{3}{24};\ \frac{1 \times 4}{8 \times 4} = \frac{4}{32};\ \frac{1 \times 5}{8 \times 5} = \frac{5}{40}$$

   The four equivalent fractions of $\frac{1}{8}$ are $\frac{2}{16}, \frac{3}{24}, \frac{4}{32}$ and $\frac{5}{40}$.

   c. The given fraction is $\frac{7}{12}$.

   Multiplying the numerator and denominator by 2, 3, 5 and 10, we get,

   $$\frac{7 \times 2}{12 \times 2} = \frac{14}{24};\ \frac{7 \times 3}{12 \times 3} = \frac{21}{36};\ \frac{7 \times 5}{12 \times 5} = \frac{35}{60};\ \frac{7 \times 10}{12 \times 10} = \frac{70}{120}$$

   The four equivalent fractions of $\frac{7}{12}$ are $\frac{14}{24}, \frac{21}{36}, \frac{35}{60}$ and $\frac{70}{120}$.

d. The given fraction is $\frac{9}{15}$.

Dividing the numerator and denominator by 3, we get,

$$\frac{9 \div 3}{15 \div 3} = \frac{3}{5}$$

Multiplying the numerator and denominator by 2, 3 and 4, we get,

$$\frac{9 \times 2}{15 \times 2} = \frac{18}{30}; \quad \frac{9 \times 3}{15 \times 3} = \frac{27}{45}; \quad \frac{9 \times 4}{15 \times 4} = \frac{36}{60}$$

Four equivalent fractions of $\frac{9}{15}$ are $\frac{3}{5}, \frac{18}{30}, \frac{27}{45}$ and $\frac{36}{60}$.

e. The given fraction is $\frac{3}{14}$.

Multiplying the numerator and denominator by 2, 3, 4 and 5, we get,

$$\frac{3 \times 2}{14 \times 2} = \frac{6}{28}; \quad \frac{3 \times 3}{14 \times 3} = \frac{9}{42}; \quad \frac{3 \times 4}{14 \times 4} = \frac{12}{56}; \quad \frac{3 \times 5}{14 \times 5} = \frac{15}{70}$$

Four equivalent fraction of $\frac{3}{14}$ are $\frac{6}{28}, \frac{9}{42}, \frac{12}{56}$ and $\frac{15}{70}$.

f. $\frac{2}{16}$

Dividing the numerator and denominator by 2, we get $\frac{2 \div 2}{16 \div 2} = \frac{1}{8}$

Multiplying the numerator and denominator by 3, 4, and 5, we get,

$$\frac{2 \times 3}{16 \times 3} = \frac{6}{48}; \quad \frac{2 \times 4}{16 \times 4} = \frac{8}{64}; \quad \frac{2 \times 5}{16 \times 5} = \frac{10}{80}$$

Four equivalent fractions of $\frac{2}{16}$ are $\frac{1}{8}, \frac{6}{48}, \frac{8}{64}$ and $\frac{10}{80}$.

6. a. $\frac{3}{8} = \frac{24}{\boxed{?}}$

8 times 3 gives 24. $3 \times 8 = 24$.

So, 8 times 8 should be the denominator. $(8 \times 8 = 64)$

$\therefore \frac{3}{8} = \frac{24}{64}$

b. $\frac{5}{7} = \frac{20}{\boxed{?}}$

4 times 5 gives 20. $5 \times 4 = 20$.

So, 4 times 7 should be the denominator. $(4 \times 7 = 28)$

$\therefore \frac{5}{7} = \frac{20}{28}$

c. $\frac{40}{44} = \frac{\boxed{?}}{11}$

44 divided by 4 gives 11, $44 \div 4 = 11$

So, $40 \div 4$, that is, 10 should be the numerator.

$\therefore \frac{40}{44} = \frac{10}{11}$

d. $\frac{28}{42} = \frac{\boxed{?}}{6}$

42 divided by 7 gives 6. $42 \div 7 = 6$

So, 28 divided by 7 should be the numerator ($28 \div 7 = 4$)

$\therefore \frac{28}{42} = \frac{4}{6}$

e. $\frac{11}{12} = \frac{\boxed{}}{48}$

4 times 12 gives 48. $12 \times 4 = 48$

So, 4 times 11 should be the numerator. ($11 \times 4 = 44$)

$\therefore \frac{11}{12} = \frac{44}{48}$

f. $\frac{27}{45} = \frac{\boxed{}}{5}$

45 divided by 9 gives 5. $45 \div 9 = 5$

So, 27 divided by 9 should be the numerator ($27 \div 9 = 3$)

$\therefore \frac{27}{45} = \frac{3}{5}$

Application Based Questions

7. **a.** $\frac{3}{5}$ and $\frac{12}{20}$

Taking cross products, we get $3 \times 20 = 60$ and $5 \times 12 = 60$**.**

So, the products are equal; hence the fractions are equivalent.

b. However, in 3/4 , $3 \times 20 = 60$, and $4 \times 12 = 48$.

Therefore, $\frac{12}{20} \neq \frac{3}{4}$

8. $\frac{4}{12}$ or $\frac{1}{3}$ fraction of the spaces has paint. $\frac{8}{12}$ or $\frac{2}{3}$ fraction of the spaces is empty.

No, these fractions are not equal as $1 \times 3 \neq 2 \times 3$.

9.
$$\frac{2}{3} = \frac{2 \times 5}{3 \times 5} = \frac{10}{15}$$
$$\frac{2}{3} = \frac{2 \times 3}{3 \times 3} = \frac{6}{9}$$
$$\frac{2}{3} = \frac{2 \times 6}{3 \times 6} = \frac{12}{18}$$
$$\frac{2}{3} = \frac{2 \times 4}{3 \times 4} = \frac{8}{12}$$
$$\frac{2}{3} = \frac{2 \times 7}{3 \times 7} = \frac{14}{21}$$

So, $\frac{2}{3}, \frac{10}{15}, \frac{12}{18}$, and $\frac{14}{21}$ are equivalent fractions.

Let's Try! 7

1. $\frac{1}{3}$ and $\frac{1}{2}$ have the some numerator. So, the fraction with smaller denominator is greater.
   As, $2 < 3$, so $\frac{1}{2}$ is greater than $\frac{1}{3}$.

   It is also clear from the above figures that , $\frac{1}{2} > \frac{1}{3}$.
2. $\frac{3}{7}$ and $\frac{4}{7}$ are like fractions. So, the fraction with greater numerator is greater.
   As $4 > 3$, so $\frac{4}{7}$ is greater than $\frac{3}{7}$.

   It is also clear from the above figures that, $\frac{4}{7} > \frac{3}{7}$.

Exercise 6E

2. **a.** $\frac{2}{5}, \frac{2}{3}, \frac{2}{9}, \frac{2}{11}, \frac{2}{7}$
   All these fractions have same numerator.
   Among the denominators, $11 > 9 > 7 > 5 > 3$, so
   $\frac{2}{11} < \frac{2}{9} < \frac{2}{7} < \frac{2}{5} < \frac{2}{3}$

   b. $\frac{3}{13}, \frac{6}{13}, \frac{5}{13}, \frac{7}{13}, \frac{2}{13}$
   All these fractions have the same denominator, hence greater the numerator, greater the fraction.

   So, $\frac{2}{13} < \frac{3}{13} < \frac{5}{13} < \frac{6}{13} < \frac{7}{13}$

   c. $1, \frac{9}{8}, \frac{3}{8}, \frac{13}{8}, \frac{5}{8}$

   In all these fractions the denominator is same that is 8. 1 means $\frac{8}{8}$.

   So, $\frac{3}{8} < \frac{5}{8} < 1 \left(\frac{8}{8}\right) < \frac{9}{8} < \frac{13}{8}$

3. a. $\frac{2}{9}, \frac{14}{9}, \frac{1}{9}, \frac{11}{9}, \frac{7}{9}$
   Since the denominator is the same, the greater the numerator, the greater the fraction.
   So, $\frac{14}{9} > \frac{11}{9} > \frac{7}{9} > \frac{2}{9} > \frac{1}{9}$

   b. $\frac{1}{3}, \frac{1}{8}, \frac{1}{2}, \frac{1}{10}, \frac{1}{6}$
   These are unit fractions so the greater the denominator, lesser the fraction.
   Hence, $\frac{1}{2} > \frac{1}{3} > \frac{1}{6} > \frac{1}{8} > \frac{1}{10}$

   c. $\frac{9}{13}, \frac{25}{13}, \frac{6}{13}, 1, \frac{14}{13}$
   These fractions are like fractions, so greater the numerator, greater the fraction.
   Hence, $\frac{25}{13} > \frac{14}{13} > 1 \left(\frac{13}{13}\right) > \frac{9}{13} > \frac{6}{13}$

Application Based Questions

4. Fraction of pizza eaten by Firoz: $\frac{3}{6}$ or $\frac{1}{2}$

   Fraction of pizza eaten by Mohit: $\frac{2}{6}$ or $\frac{1}{3}$

   As $\frac{1}{2}$ is bigger than $\frac{1}{3}$, Firoz ate more of the pizza.

   Ans: Firoz ate more pizza than Mohit

5. Portion of book read on Friday: $\frac{2}{7}$

   Portion of book read on Sunday: $\frac{2}{5}$

   As $\frac{2}{5} > \frac{2}{7}$, Tara read more on Sunday.

   Ans: Tara read more on Sunday.

Let's Try 8

1. $\frac{1}{9} + \frac{3}{9} = \frac{4}{9}$

2. $\frac{7}{11} - \frac{2}{11} = \frac{5}{11}$

Exercise 6F

2. **a.** $\frac{5}{8} + \frac{3}{8} = \frac{5+3}{8} = \frac{8}{8} = 1$

   b. $\frac{2}{9} + \frac{3}{9} = \frac{2+3}{9} = \frac{5}{9}$

   c. $\frac{1}{7} + \frac{3}{7} = \frac{1+3}{7} = \frac{4}{7}$

   d. $\frac{3}{13} + \frac{5}{13} = \frac{3+5}{13} = \frac{8}{13}$

   e. $1 + \frac{1}{5} = \frac{5}{5} + \frac{1}{5} = \frac{5+1}{5} = \frac{6}{5} = 1\frac{1}{5}$

3. **a.** $\frac{9}{10} - \frac{4}{10} = \frac{9-4}{10} = \frac{5}{10} = \frac{1}{2}$

   b. $\frac{11}{12} - \frac{5}{12} = \frac{11-5}{12} = \frac{6}{12} = \frac{1}{2}$

   c. $\frac{13}{15} - \frac{7}{15} = \frac{13-7}{15} = \frac{6}{15} = \frac{2}{5}$

   d. $\frac{11}{20} - \frac{9}{20} = \frac{11-9}{20} = \frac{2}{20} = \frac{1}{10}$

e. $\mathbf{1 - \frac{1}{3} = \frac{3}{3} - \frac{1}{3} = \frac{3-1}{3} = \frac{2}{3}}$

Application Based Questions

4. Fraction of rangoli done on Monday: $\frac{3}{10}$

Fraction of rangoli done on Tuesday: $\frac{1}{10}$

Fraction of rangoli done on Wednesday: $\frac{5}{10}$

Total rangoli completed in 3 days: $\frac{3}{10} + \frac{1}{10} + \frac{5}{10}$

$= \frac{3+1+5}{10} = \frac{9}{10}$

Ans: Sujata completed 9/10 of the rangoli in three days.

5. Oil consumed in first week: $\frac{3}{8}$

Oil consumed in second week: $\frac{2}{8}$

Total oil consumed = $\frac{3}{8} + \frac{2}{8}$

$= \frac{3+2}{8} = \frac{5}{8}$

Quantity of oil left over = $1 - \frac{5}{8}$

$= \frac{8}{8} - \frac{5}{8} = \frac{8-5}{8}$

$= \frac{3}{8}$

Ans: Rekha consumed a total of $\frac{5}{8}$ fraction of oil in two weeks and $\frac{3}{8}$ fraction of oil was left.

3. Raju's share of cake: $\frac{3}{8}$

Meenu's share of cake: $\frac{1}{8}$

As $\frac{3}{8} > \frac{1}{8}$, Raju's share is bigger.

Ans: Raju's share of cake is bigger than Meenu's.

4. Share of flour: $\frac{3}{8}$

Share of milk powder: $\frac{2}{8}$

Share of sugar: $\frac{1}{8}$

Quantity of water = 1 – (share of flour + share of milk powder + share of sugar)

$= 1 - (\frac{3}{8} + \frac{2}{8} + \frac{1}{8})$

$= 1 - (\frac{3+2+1}{8})$

$= 1 - (\frac{5}{8})$

$= \frac{8-6}{8}$

$= \frac{2}{8} = \frac{1}{4}$

Ans: a. The fraction of water used was $\frac{1}{4}$.

b. Among the ingredients flour was used more.

5. Fraction of salary for medical expenses: $\frac{1}{10}$

Fraction of salary for house rent: $\frac{2}{10}$

Fraction of salary for food: $\frac{4}{10}$

Fraction of rest of salary = $1 - (\frac{1}{10} + \frac{2}{10} + \frac{4}{10})$

$= 1 - (\frac{1+2+4}{10})$

$= 1 - \frac{7}{10}$

$= \frac{10-7}{10} = \frac{3}{10}$

Ans: The fraction of rest of the salary is $\frac{3}{10}$.

Revision Exercises

1. Identify the proper fractions among the following.

a. $\frac{3}{7}$ proper fraction **b.** $\frac{9}{8}$ improper fraction

c. $\frac{1}{5}$ proper fraction **d.** $\frac{7}{13}$ proper fraction

2. Change the improper fractions to mixed numbers.

a. $\frac{56}{3} = 18\frac{2}{3}$ **b.** $\frac{49}{10} = 4\frac{9}{10}$ **c.** $\frac{68}{7} = 9\frac{5}{7}$

d. $\frac{51}{8} = 6\frac{3}{8}$ **e.** $\frac{80}{7} = 11\frac{3}{7}$

3. Write the mixed fractions as improper fractions.

a. $2\frac{3}{7} = \frac{17}{7}$ **b.** $5\frac{9}{10} = \frac{59}{10}$ **c.** $4\frac{5}{8} = \frac{37}{8}$

d. $7\frac{1}{6} = \frac{43}{6}$ **e.** $8\frac{2}{5} = \frac{42}{5}$

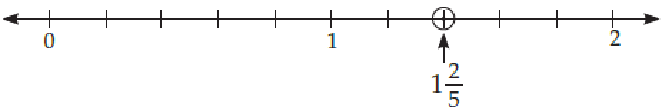
4. Write the simplest form of the given fractions

a. $\frac{45}{60} = \frac{3}{4}$ **b.** $\frac{42}{66} = \frac{7}{11}$ **c.** $\frac{38}{76} = \frac{1}{2}$

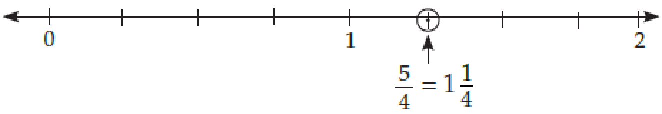
d. $\frac{24}{108} = \frac{2}{9}$ **e.** $\frac{81}{99} = \frac{9}{11}$

5.

a. $\frac{7}{5} = 1\frac{2}{5}$



b. $\frac{5}{4} = 1\frac{1}{4}$



6. a. $\frac{2}{10}$ b. $\frac{3}{8}$

Level Up

1. Find the sum or difference. Simplify your answer if possible.

a. $\frac{3}{15} + \frac{4}{15} + \frac{1}{15}$

$= \frac{3+4+1}{15}$

$= \frac{8}{15}$ **(Ans)**

b. $\frac{7}{20} + \frac{6}{20} + \frac{2}{20}$

$= \frac{7+6+2}{20} = \frac{15}{20}$

$= \frac{15}{20} = \frac{3}{4}$ **(Ans)**

c. $\frac{9}{14} - \frac{5}{14}$

$= \frac{9-5}{14} = \frac{4}{14}$

$= \frac{4}{14} = \frac{2}{7}$ **(Ans)**

d. $\frac{22}{25} - \frac{9}{25}$

$= \frac{22-9}{25} = \frac{13}{25}$

$= \frac{13}{25}$ **(Ans)**

e. $1 - \frac{7}{10} = \frac{10-7}{10} = \frac{3}{10}$

f. $\frac{7}{8} + \frac{5}{8} = \frac{7+5}{8} = \frac{12}{8} = \frac{3}{2} = 1\frac{1}{2}$

2. Solve the following word problems

a. Fraction of glass with fruit juice concentrate: $\frac{1}{4}$

Fraction of glass with water: $\frac{2}{4}$

Part of glass filled = $\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$

Part of the glass which is empty: $1 - \frac{3}{4} = \frac{4-3}{4} = \frac{1}{4}$

Ans: Three fourth of the glass is filled and one fourth is empty.

b. Portion of wall painted by Ravi = $\frac{3}{8}$

Portion of wall painted by Chirag = $\frac{1}{8}$

Total wall painted on first day = $\frac{3}{8} + \frac{1}{8} = \frac{4}{8} = \frac{4}{8} = \frac{1}{2}$

Portion of the wall to be completed on second day = $1 - \frac{1}{2} = \frac{2-1}{2} = \frac{1}{2}$

Ans: Half of the wall remained to be painted on the second day.

c. Total evening time of Bobby: 6 hours

Fraction of the time on studying: $\frac{2}{6} = \frac{1}{3}$

Fraction of the time on playing: $\frac{2}{6} = \frac{1}{3}$

Fraction of time doing chores: $\frac{1}{6}$

Fraction of time spent at dinner: $\frac{1}{6}$

d. Number of trees in the botanical garden = 78

Number of shrubs = 92

Number of climbers = 45

Number of herbs = 65

Total number of plants in the garden = 280

i) Fraction of trees in the garden = $\frac{78}{280}$

Fraction of shrubs in the garden = $\frac{92}{280}$

Fraction of climbers in the garden = $\frac{45}{280}$

Fraction of herbs in the garden = $\frac{65}{280}$

ii) Fraction of shrubs and herbal plants = $\frac{157}{280}$

3. Total number of students = 32

a. Number of students who participated in 'Plant a tree' activity = one fourth of 32 = 32 ÷ 4 = 8.

b. Fraction of students who participated in 'Shop wisely' activity = $\frac{2}{4} = \frac{1}{2}$

Fraction of students participated in 'bike more drive less' = $1 - (\frac{1}{4} + \frac{2}{4})$

$$1 - \left(\frac{1+2}{4}\right) = 1 - \frac{3}{4} = \frac{4-3}{4} = \frac{1}{4}$$

Ans: a. 8 students participated in Plant a tree activity.

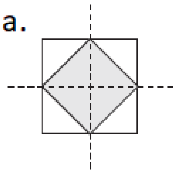
b. $\frac{1}{2}$ the students participated in shop wisely and $\frac{1}{4}$ students took part in 'bike more drive less' activity.

c. We can save the environment by planting more trees, using public transport running on green fuels and renewable energy sources like battery operated vehicles.

Hots Questions

1.

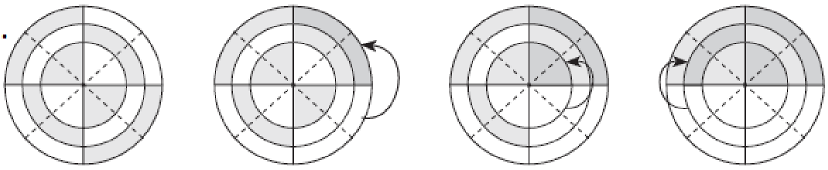
1. a.



Total parts = 8, shaded parts = 4

$\therefore$ Fraction of shaded region = $\frac{4}{8} = \frac{1}{2}$

b.



We remove the colour from one part at a time and colour another equal part as shown in the above figures.

In the fourth figure we see that $\frac{4}{8}$ of the figure is shaded.

$\therefore$ Fraction of shaded region = $\frac{4}{8} = \frac{4 \div 4}{8 \div 4} = \frac{1}{2}$

2. One third of a group of children may not be equal to one third of another group of children as the total number of children in the two groups might be different. For example, if one group has nine children, then one third of nine will be three. But if another group has 15 children, then one third of that group of children will be five.

Everyday Maths

1. a. Quantity of black mustard seeds and chana dal = $\frac{3}{2} + \frac{1}{2}$ teaspoons

$= \frac{3+1}{2} = \frac{4}{2} = 2$ teaspoons.

b. The quantity of long grain rice $1\frac{1}{2} = \frac{3}{2}$ cups.

c. Out of rice and urad dal, rice is used more.

Difference between quantities of ghee and black mustard seeds = $4 - \frac{3}{2}$ teaspoons

$= \frac{8-3}{2} = \frac{5}{2}$ teaspoons

$= \frac{5}{2} = 2\frac{1}{2}$ teaspoons